

plates, alleviating the need for magnetic flux guiding and shielding components that add bulk and cost to inductive solutions. The realisable amount of coupling capacitance is limited by the available area of the device, imposing a design constraint on contactless power delivery.

Parallel-plate capacitance across a 1/4mm air gap is only 3.5 pF/cm², limiting typical interface capacitance to a few tens of picofarads. The required charging power is upwards of 2.5W (USB-specification). Existing capacitive power transfer (CPT) solutions either use much larger capacitors or are targeted at lower-power applications, such as coupling of power and data between integrated circuits, or transmission of power and data to biosignal instrumentation systems.

Murata's capacitive-coupling wireless power transmission modules have two sets of asymmetric dipoles consisting of active and passive electrodes positioned vertically on the power transmitting and receiving sides. Power is transmitted utilising an induction field generated by coupling of the two sets of asymmetric dipoles. This configuration realises wireless power transmission with high position freedom and efficiency.

Applications

Wireless charging of consumer electronics. Automatic wireless charging of mobile electronics (laptops, phones, game controllers, etc) in homes, cars, offices and Wi-Fi hotspots is possible. Direct wireless powering of stationary devices is also facilitated, thus eliminating custom wiring, wall-wart power supplies and unsightly cables.

Wireless charging is done using an electromagnetic field to transfer energy between two objects via a charging station. Energy is sent through inductive

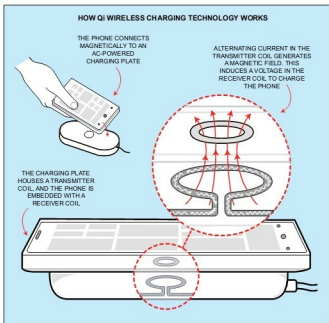


Fig. 4: Qi standard

coupling to an electrical device, which then uses that energy to charge batteries or run the device.

Wireless chargers typically use an induction coil to create an alternating electromagnetic field from within a charging base station. Another induction coil in the portable device takes power from the electromagnetic field and converts it back into electrical current to charge the battery.

Wirelessly charging of electrical vehicles. Researchers at Korea Advanced Institute of Science and Technology (KAIST) have developed an electric transport system, named the Online Electric Vehicle, OLEV, in which the vehicles get their power from cables underneath the surface of the road via non-contact magnetic charging. The power source is placed underneath the road surface and power is wirelessly picked up by the vehicle itself. Once logistics is worked out for this idea, the day isn't far off when charging vehicles without having to stop will become a reality.

The Qi standard

Qi (pronounced 'chee') is an interface standard developed by the Wireless Power Consortium for wireless electrical power transfer over distances of up

to 40 millimetres (1.6 inches). The Qi system comprises a power transmission pad and a compatible receiver in a portable device. The mobile device is placed on top of the power transmission pad, which charges the device via magnetic induction.

Mobile device manufacturers that are complying to the standard include HTC, LG Electronics, Huawei, Motorola Mobility, Samsung, Nokia and Sony. They aim to create a global standard for wireless charging technology.

All in all

In this era of communications and connectivity, individuals have multiple technologies to support their day-to-day requirements. In this scenario of multiple gadgets, wireless power technology is emerging as a practical solution for portable devices. The predominant market for wireless power transfer is unarguably the consumer handheld market. Mobile equipment giants like Nokia, LG, Samsung, HTC and Motorola have already become WPC Qi-compliant and offer compatible phones.

Pavan Pudipeddi, senior strategic marketing manager, Wireless Power and Battery Charge Management, Texas Instruments (TI), says, "At TI, we recognise the huge market available in the wireless energy domain. TI's goal is to become the premier supplier of wireless power solutions in the industry. Our receiver chips integrate a low-impedance, fully synchronous rectifier along with a low-dropout regulator, digital control, and accurate voltage and current loops. Recently, TI's bq51050B became the industry's first Wireless Power Consortium (WPC) 1.1 Qi-compliant wireless power receiver with integrated direct battery charger. It enables a faster, more efficient way of charging smartphones, wireless keyboards and other portable electronics." ●

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